MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

DSC – MG2DSCCFS100 – INTRODUCTION TO ETHICAL HACKING AND TOOLS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

PART A

Answer any **10** questions

Each question carries **1** mark

1. Explain about Grey Hat Hacker. [U]/[CO1]
   Grey hat refers to an individual who operates without malicious intent but still violates laws or acceptable standards of behavior by exploiting security vulnerabilities without permission. Grey hat hackers exist between white hat and black hat hacking.
   Explanation – 1 mark

2. Discuss about the different types of hackers. [U]/[CO1]
   White Hat, black hat, grey hat, blue hat, red hat, green hat, script kiddies, hacktivists, state/nation sponsored hackers etc.
   Identifying different types, brief explanation – 1 mark

3. Compare hacking and ethical hacking. [U]/[CO1]
   Hacking is unauthorized intrusion into computer systems and networks, often with malicious intent. Ethical Hacking, on the other hand, involves similar techniques but is conducted with legal authorization and aims to identify and rectify security vulnerabilities. While hacking is criminal, illegitimate and unfavourable, ethical hacking is legal and favorable to enhance security.
   Comparison – 1 mark.

4. Describe the term “Vulnerability”. [U]/[CO1]
   A vulnerability is a weakness in an IT system that can be exploited by an attacker to deliver a successful attack. They can occur through flaws, bad coding/features, user error or a combination of these.
   Explanation – 1 mark

5. Explain cyber attack. [U]/[CO1]
   Cyber attacks are deliberate attempts by threat actors to gain unauthorized access to systems, steal sensitive data, disrupt operations, or damage digital infrastructure. These attacks can target individuals, businesses, or governments and often exploit weaknesses in software, human behavior, or system configurations.
   Explanation – 1 mark

6. Explain Vulnerability. [U]/[CO1,CO2]
   A vulnerability is a weakness in an IT system that can be exploited by an attacker to deliver a successful attack. They can occur through flaws, bad coding/features, user error or a combination of these.
   Explanation – 1 mark

7. Explain Malware attacks. [U]/[CO1,CO2]
   A malware attack is a cyberattack where malware (malicious software) executes unauthorized actions on the victim's system. The malicious software encompasses many specific types of attacks such as ransomware, spyware, command and control, and more.
   Definition – ½, explanation – ½

8. Explain input validation attacks. [U]/[CO1,CO2]
   Input validation attacks are a method of cyberattack in which the attacker injects malicious input that can be interpreted and executed by a target system to exploit its vulnerabilities.
   Definition – ½, explanation – ½

9. Explain botnets. [U]/[CO2]
   A botnet is a network of infected devices (bots) controlled remotely by an attacker (botmaster) through a Command and Control (C&C) system.
   Explanation – 1 mark

10. Identify the network layer protocols. [U]/[CO1]
    IP, ARP, ICMP, IGMP
    Identify – ½, brief explanation – ½

11. Compare hardware firewall and software firewall. [An]/[CO3]
    A software firewall is a special type of software that runs on a computer/server to secure it malicious attempts. A hardware firewall is a dedicated appliance. It has its own processors, memory, and interfaces built solely for inspecting and controlling traffic.
    Comparison – 1 mark

12. Identify an example for Hardware Firewall. [U]/[CO1]

    Cisco ASA, SonicWall Network Security Appliances, PaloAlto Networks series etc.

    Identify – ½, brief explanation – ½

[1x10=10]

PART B

Answer any **4** questions

Each question carries **5** marks

13. Discuss about the purpose of maintaining access to a compromised system. [U]/[CO1]

    The purpose of maintaining access to a compromised system (often called persistence) is to ensure that an attacker—or, in ethical hacking, a security tester—can continue to interact with the system even after the initial breach – continuous control, monitor target activities, achieve privilege escalation, launching further attacks etc.

    Maintaining access definition – 1 mark

    Purpose explanation – 4 marks

14. Describe the different techniques used to clear tracks. [U]/[CO1]

    "Clearing tracks" refers to actions taken after a compromise to hide evidence of unauthorized activity and avoid detection. The various techniques invlolve Log Manipulation or Deletion, File and Artifact Removal, Timestamp Manipulation (Timestomping), Clearing Command-line History, Use of Anti-Forensic Tools, Rootkits and Hidden Processes, Disabling Security Mechanisms, Obfuscation and Encryption, Clearing Network Traces etc.

    Definition – 1 mark

    Identifying and explaining different techniques – 4 marks

15. Discuss about Man-In-The-Middle attacks. [U]/[CO1,CO2]

    A man-in-the-middle (MITM) attack is a cyberattack in which a hacker steals sensitive information by eavesdropping on communications between two online targets. Instead of User ↔ Server, MITM uses User ↔ Attacker ↔ Server.

    Stages in MITM attacks – Interception, Decryption, Manipulation, Relaying.

    Common Techniques Used in MITM - ARP Spoofing, DNS Spoofing, Session Hijacking etc.

Definition – 1 mark

Stages – 1 mark

Common Techniques – 1 mark

Detection methods – 1 mark

Prevention methods – 1 mark

16. Explain Buffer overflow attacks. List out some consequences of buffer overflow attacks. [U]/[CO1,CO2]

    A buffer overflow attack occurs when a program writes more data into a memory buffer than it can hold. Because buffers have fixed sizes, extra data "overflows" into adjacent memory, potentially overwriting critical information.

    Consequences - Arbitrary Code Execution, System Crashes, Privilege Escalation, Data Corruption, Unauthorized Access, Malware Injection, Compromise of Entire System etc.

    Definition – 1 mark

    Consequences - 4 marks

17. Compare about any 5 network protocols. [An]/[CO3]

    IP, DNS, HTTP, HTTPS, DHCP, FTP etc. – explain full form, purpose, port number, security etc.

    5 Protocols and explanation – 1 mark each

18. Discuss about the methods used to protect websites. [U]/[CO1]

    Use Secure Communication (HTTPS), Input Validation and Sanitization (to prevent SQL injection, XSS etc.), Strong Authentication & Authorization (use of strong passwords, MFA), Web Application Firewall, Regular Updates and Patch Management, Secure Session Management, Regular Security Testing, Backup and Recovery Plan, Logging and Monitoring, Secure Configuration.

    Any 5 methods and explanation – 1 mark each

[5x4=20]

PART C

Answer any **2** questions

Each question carries **20** marks

19. Describe the process of authorization and explain the different types of authorization techniques. [U]/[CO1]

Authorization is the process of determining what actions an authenticated user or service is allowed to perform.

Types of Authorization Techniques - Discretionary Access Control (DAC), Mandatory Access Control (MAC), Role-Based Access Control (RBAC), Attribute-Based Access Control (ABAC), Rule-Based Access Control, Access Control Lists (ACLs).

Definition – 2 marks

Types identification – 2 marks

Types explanation – Any 4 types – 4 marks each (16 marks)

20. Discuss in detail about Social engineering attacks. [U]/[CO1,CO2]

Social engineering attacks are techniques in which attackers manipulate people into revealing confidential information or performing actions that compromise security.

Explain common types of social engineering attacks, consequences, prevention.

Definition – 1 mark

Identification of different types – 1 mark

Explanation(attack, consequences, prevention) – explain any 3 attacks – 6 marks each(18 marks)

21. Differentiate between HIDS and NIDS. [An]/[CO3]

HIDS (Host-Based Intrusion Detection System) and NIDS (Network-Based Intrusion Detection System) are security tools used to detect malicious activities, but they differ in where and how they monitor.

HIDS is installed on individual systems (hosts) to monitor internal activities like file changes, logs, and processes.

NIDS is placed on a network to monitor traffic flowing between devices and detect suspicious patterns.

Differentiate in terms of location, monitoring, scope, detection, performance, deployment etc.

HIDS definition – 2 marks, explanation – 8 marks

NIDS definition – 2 marks, explanation – 8 marks

[20x2=40]